

DATA 227 — Autumn 2026 | Homework 1

Data Visualization and Communication

Trimble

Due: Tuesday, October 6, 2026, 11:59 pm

Reminder: the exercises are mostly practice, so that you can produce something to show off your skills by the end of the quarter.

Submission. Upload your responses to Gradescope as a PDF or HTML file, and put your answers to both questions in a single file. Attach the most relevant pieces of your code in any convenient format. If your answers are not in PDF or HTML, we will send your homework back ungraded.

- 1. Population pyramid.** Create a static visualization of the population histograms from the 1900 and 2000 U.S. Census. The data come from the U.S. Census Bureau via IPUMS (<https://www.ipums.org/>) and give the number of people enumerated in 5-year age bins (0–4, 5–9, 10–14, ..., up to 90+), by reported sex (1 for male, 2 for female). The file is `data/census1900-2000.csv`, available on Canvas. It looks like this:

```
Sex,Year,Age,People
```

```
1,1900,0,4619544
```

```
1,2000,0,9735380
```

```
1,1900,5,4465783
```

```
1,2000,5,10552146
```

```
...
```

You can read about this data, along with some light commentary on age histograms in the Gilded Age, in *1900 Census: Volume II. Population, Part 2, Statistics of the Population*:

<http://www2.census.gov/library/publications/decennial/1900/volume-2/volume-2-p2.pdf>

Examine the data, ask a question of the data, and design a visualization that answers the question.

Include a figure caption, at most one paragraph describing your findings, and any code you used to produce the visualization. You do not have to use any specific tools, but you should find an interesting question and answer it effectively.

- 2. One-, two-, and three-variable graphs from the 2006 CDC birth data.** The U.S. CDC has collected a handful of fields on nearly every birth in the United States for 50 years. We will look at births in 2006, available at

https://www.cdc.gov/nchs/data_access/Vitalstatsonline.htm

You will want `Nat2006US.zip` (a text file with 4 million lines) and `UserGuide2006.pdf` (a data dictionary giving the fields, their positions, and the meaning of all the symbols used).

Make three static visualizations that communicate, respectively, the distribution of one variable, the relationship between two variables, and the relationship between three variables in the 2006 CDC birth data. Use an appropriate graph type for each, and write a caption for each graph. Put all three graphs on one page.

The 2006 data include the following fields: day of the week, month of the year, age of mother, age of father (most of the time; this was a survey), country of mother's birth, country of mother's residence, self-reported race and Hispanic status, education level, whether the birth was one of multiple births, child number, questions about maternal smoking habits, duration of pregnancy, and whether the delivery was vaginal or surgical.

The dataset may require some effort to decode. I will give you some starter code that may help you extract interesting fields.

Include the essential piece of code that you use to generate your visualizations (instructional staff won't try to run your code but we'd like to glance at it).

Include a statement describing whether and how you used generative AI in preparing your homework.